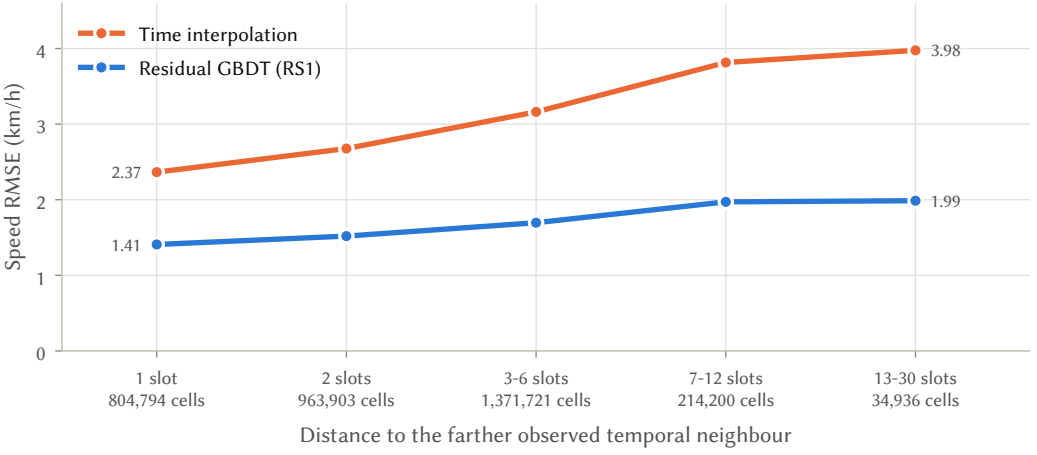The learned residual halves interpolation error at every gap length